WOW Factor: Tourism Transport Disruption and Destination Impacts in Iceland

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Article Title
WOW Factor: Tourism Transport Disruption and Destination Impacts in Iceland

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Abstract

This study estimates how the sudden loss of low-cost air service affected visitor flows and labor-market outcomes in Iceland after WOW Air's 2019 bankruptcy. A Bayesian structural time-series model constructs counterfactual paths for Iceland as if the airline had remained in operation. Passenger, employment, unemployment, and wage outcomes are projected using Iceland's pre-bankruptcy trajectory and monthly indicators from nearby Nordic and North Atlantic economies. International passenger departures were 20.6% lower than the counterfactual, equivalent to about 534,000 fewer passengers, with foreign passengers accounting for over 80% of the decline. **Passenger losses were uneven across nationalities, with some of the largest shortfalls among travelers from markets heavily served by WOW Air. Over the same period,** air-transport employment was 25.9% below expected levels, **while tourism employment net of air transport was 6.6% lower. The results show a substantial decline in foreign passenger flows following the airline's exit, accompanied by measurable short-run deterioration in tourism-facing labor markets.**

Keywords: tourism shocks; destination vulnerability; transport connectivity; island destinations; tourism labor markets; tourism planning;

1 Introduction

Air service is central to tourism, particularly where travel alternatives are limited. Flight availability, markets served, and fares influence both who visits and how many travelers can reach a destination. This dependence is especially pronounced for island destinations, where tourism relies heavily on organized air and sea transport and substitutes may be limited or unavailable. Hotels, restaurants, tour operators, and their workers are dependent on the upstream flow of visitors. The bankruptcy of Icelandic airline WOW Air in 2019 provides a clear case of this risk in Ice-

land, where the collapse of a low-cost carrier sharply reduced affordable international travel options.

WOW Air was a budget airline launched in 2012 that became well-known for its offering of $99 flights from the United States and Europe to Iceland. The low-cost carrier grew rapidly, enabling and benefiting from a general tourism boom in Iceland. By 2018, the carrier employed over 1,000 persons[1] and held over a 30%[2] share of the seats to and from Iceland. However, it ceased operations abruptly in early 2019, leaving passengers stranded across the United States, Iceland, and Europe.

Tourism planners often focus on making destinations attractive, but realized tourism also depends on a complete supply chain to translate potential demand into actual visits. **The WOW Air case provides a useful shock for plausibly differentiating a reduction in tourism flows associated with the loss of air service from a broader decline in tourism demand.** WOW Air's bankruptcy reduced low-cost travel options despite Iceland remaining an attractive destination, and global tourism was generally undisrupted. This setting allows for estimation of how much realized visitor traffic and tourism-facing employment depended on connectivity, holding other conditions fairly constant.

Iceland is a remote island destination with few practical substitutes for air travel, making it a useful setting for studying how air-service disruptions move through a tourism labor market. International visitors overwhelmingly arrive by air. In 2019, 98.7% of foreign visitors came through Keflavík International Airport, while about 18,100 arrived by ferry.[3] **That reliance makes airline service a central part of Iceland's tourism economy. WOW Air's collapse therefore provides a well-bounded case for estimating how the sudden loss of airline routes and low-cost travel options affected passenger flows**

[1] See CNBC, "Icelandic airline WOW Air collapses and cancels all flights," https://www.cnbc.com/2019/03/28/icelandic-airline-wow-air-collapses-and-cancels-all-flights.html.

[2] See Simple Flying, "Only Room For One: How Icelandair Can Pick Up WOW Air's Capacity," https://simpleflying.com/only-room-for-one-how-icelandair-can-pick-up-wow-airs-capacity/.

[3] See Icelandic Tourist Board (Ferðamálastofa), "Total number of tourists just over 2 million in 2019," https://www.ferdamalastofa.is/is/um-ferdamalastofu/frettir/riflega-2-milljonir-ferdamanna-2019.

and the tourism labor market. The case presented is also replicable and remains relevant for tourism planning. Airline exits are not rare historical events, and **tourism growth that depends heavily on particular carriers or routes may create risks if that service is withdrawn** (Bachwich and Wittman, 2017; Sun et al., 2025).

Tourism can be a boon to local economies, generating employment, foreign exchange, and regional growth (Gwenhure and Odhiambo, 2017; Seetanah, 2011; Faber and Gaubert, 2019). These benefits are especially important for small island economies with limited alternative export sectors (Seetanah, 2011; Roudi, Arasli and Akadiri, 2018; Kumar and Stauvermann, 2021). **Low-cost carriers can contribute substantially to tourism growth by lowering fares and opening new markets (Pulina and Cortés-Jiménez, 2010; Chung and Whang, 2011; Hong Tsui, 2017). At the same time, routes may be unstable and airline cessation is not uncommon, creating risks for destinations whose visitor growth depends heavily on particular carriers or routes** (Bachwich and Wittman, 2017; Sun et al., 2025). **This literature motivates the central planning concern in this paper: tourism growth can create economic benefits while also increasing the consequences of disruptions to the transportation systems that support those visitor flows.**

COVID-19 highlighted the associated risks of tourism dependence. Pandemic studies show that tourism-dependent regions and workers experienced disproportionate losses, with impacts varying by sector, region, gender, age, education, and labor-market position (Duro et al., 2021; Sun et al., 2021, 2022; Zhang and Yang, 2026). This literature is useful for showing the scale of tourism labor-market exposure, but the pandemic bundled several mechanisms at once, making it hard to isolate the consequences of **reduced visitor flows from changes in safety, policy, mobility, and broader economic conditions.** For example, closely related evidence on changing tourism employment comes from Perles-Ribes et al. (2023), who estimate the immediate impact of COVID-19 on tourism employment in Spain using ARIMA and Bayesian structural time-series methods as well. They find sub-

stantial employment losses when tourism is compared with its own pre-pandemic trajectory, but the tourism-specific effect is no longer statistically significant when the contemporaneous behavior of the rest of the economy is used as the counterfactual. This paper implements a similar counterfactual logic but examines a narrower disruption, **the sudden loss of airline service in a destination that remained otherwise open to visitors.**

Evidence on whether tourism shocks are temporary or persistent is mixed. Some studies find that arrivals or receipts recover fairly quickly after crises, especially when shocks are temporary, event-specific, or tied to conditions that can improve over time (Saleh, Verma and Ihalanayake, 2011; Yucel, 2020; Morakabati, 2019). Other evidence shows that shocks to tourism expenditures and receipts can be persistent and may require policy response rather than passive recovery (Payne and Nazlioglu, 2023; Kocak, Bulut and Bakir, 2026). **This variation makes the nature of the disruption important for tourism planning. The case study examined differs from many tourism shocks because it directly removes routes and travel options rather than primarily changing destination appeal or traveler preferences.**

Where present, low-cost carriers are **particularly relevant to this type of transportation risk.** They expand tourism by lowering fares, **adding routes and flight options**, stimulating new demand, and opening new markets (Barrett, 2004; Pulina and Cortés-Jiménez, 2010; Chung and Whang, 2011; Hong Tsui, 2017). Their strategy differs from that of full-service carriers, with greater emphasis on cost discipline, fast turnarounds, route experimentation, and price-sensitive leisure traffic (Barrett, 2004; Graham, 2013). Recent evidence on ultra-low-cost carriers, a sub-classification to which WOW Air is often assigned, highlights the tradeoff between growth and instability. In one study, 26% of new ultra-low-cost carrier markets were abandoned within two years (Bachwich and Wittman, 2017). Airline closure is also a recurring feature of aviation markets, with more than 200 airline cessations documented across 95 countries between 2010 and 2023 (Sun et al., 2025). Low-fare carriers can expand a destination's visitor market, but **tourism growth built**

around particular carriers and routes may be difficult to sustain if that service disappears and is not quickly replaced.

This paper asks two related questions. First, **does the loss of a major provider of airline service and low-cost travel options reduce passenger flows to a tourism destination? Second, when airline service is disrupted, are the effects contained within the airline sector, or do they appear more broadly across tourism-facing industries? Together, these questions examine how dependence on international tourism and air service can translate into measurable economic risk when a major carrier exits.**

This paper makes **two primary contributions** to tourism planning and development research. **First, it connects research on tourism dependence with work on low-cost carriers to examine a practical planning risk. Visitor flows and the jobs they support may depend in part on service provided by a major airline. Prior studies show that reliance on international tourism can increase economic vulnerability (Nguyen and Su, 2022; Canh and Thanh, 2020; Wang, Han and Ma, 2022), while low-cost carriers can stimulate demand and open new markets** (Pulina and Cortés-Jiménez, 2010; Chung and Whang, 2011; Hong Tsui, 2017; Santos and Cincera, 2018). **The WOW Air case connects these insights by illustrating how growth in visitor markets supported by a major carrier can leave tourism-facing firms and workers exposed when that air service disappears.**

Second, the paper provides counterfactual-backed evidence on a tourism shock that is narrower than many others studied in the literature. Pandemic, terrorism, political, natural-disaster, and climate-related shocks are often used to study tourism disruption (Saleh, Verma and Ihalanayake, 2011; Yucel, 2020; Morakabati, 2019; Duro et al., 2021; Sun et al., 2021, 2022; McCarroll, LaVanchy and Kerwin, 2024). WOW Air's bankruptcy **primarily removed airline routes and service from an island destination without closing the destination, and without other incidents otherwise discouraging**

visitation. Since the event primarily affected one destination, this paper uses a Bayesian structural time-series design with information from nearby countries to estimate the counterfactual path of passenger and labor-market outcomes.

The remainder of the paper proceeds as follows. Section 2 describes the empirical setting, data, and Bayesian structural time-series design. Section 3 reports the estimated effects on passenger flows and labor-market outcomes. Section 4 summarizes the study's limitations, and Section 5 discusses implications for tourism dependence, air-service risk, and tourism labor-market planning.

2 Materials and Methods

2.1 Empirical Setting

WOW Air began operations in 2012 and expanded rapidly alongside Iceland's tourism boom.[4] **By connecting Europe and North America through Keflavík International Airport, the airline expanded low-cost travel options to Iceland and supported continued visitor growth. Keflavík served as the hub in WOW Air's hub-and-spoke network, encouraging transatlantic passengers to stop in Iceland and add a short visit to their itinerary, often turning connecting traffic into local tourism demand.** Its March 2019 bankruptcy occurred in a setting where tourism had become increasingly central to the Icelandic economy. Between 2010 and 2018, monthly foreign visitors per worker rose from 0.24 to 0.99, while tourism employment increased from 7.88% to 14.84% of total employment. **This specialization is consequential as tourism growth creates jobs while also increasing reliance on visitor spending.**

This study examines the WOW Air bankruptcy in two steps. **First, it estimates whether the bankruptcy produced a measurable passenger gap between observed**

[4]See PBS, "What WOW's collapse means for budget airlines and tourism," `https://www.pbs.org/newshour/economy/making-sense/what-wows-collapse-means-for-budget-airlines-and-tourism`.

international travel and the travel that would have happened if the airline had not collapsed, represented by the synthetic counterfactual. Second, it examines whether that passenger gap was accompanied by broader labor-market effects, providing evidence on how the economic risks associated with tourism dependence may be realized in tourism-facing industries. The setting is quasi-experimental, with the bankruptcy providing a plausibly exogenous air-service shock that allows the analysis to examine whether reduced visitor flows were followed by broader labor-market effects.

The outcomes examined reflect this two-step empirical focus. **International passenger departures provide a consistent and reliable administrative measure of the shock to visitor flows associated with the bankruptcy**, while employment, unemployment, and taxable wages measure whether that shock reached workers and wages. **Passenger departures are used as a proxy for visitor flows rather than a direct measure of tourism because the data do not identify trip purpose. The analysis primarily uses total passenger departures, with additional nationality-level disaggregation to examine whether declines differed across passenger markets.**[5]

2.2 Data and Comparison Units

The study assembles monthly indicators from national statistical agencies in Iceland, the Faroe Islands, Greenland, Denmark, and Norway.[6] Table 1 reports the rationale for the comparison countries whose time series form the pool of covariates for the counterfactual. Each row describes one comparison criterion, including whether the country is considered Nordic, its distance to Iceland, the degree of seasonality, population size, tourism intensity,

[5]Departures are used rather than arrivals because departures are counted as travelers pass through international security after their time in Iceland. In a setting with stopover traffic and short visitor stays, departures provide a clearer measure of passengers who were plausibly present in Iceland than arrivals, which can include travelers whose connection through Keflavík did not translate into an Iceland visit.

[6]The principal statistical sources are Statistics Iceland, Statistics Faroe Islands, Statistics Greenland, Statistics Denmark, Statistics Norway, and Nordic Statistics. Some supplementary indicators are drawn from the World Bank Open Data Portal and The Global Economy where monthly or comparable official series were unavailable.

tourism growth, and other export composition. Greenland, the Faroe Islands, Denmark, and Norway form the preferred comparison set because they are nearby North Atlantic or Nordic economies, share relevant seasonal patterns, and provide monthly indicators that help predict regional economic conditions. Sweden, Ireland, and the United Kingdom are shown for transparency but are not used in the preferred model because they are larger, more diversified, or less comparable as counterfactual predictors.

The assembled dataset includes measures of passenger travel, hotel activity, overnight stays, unemployment, employment, wages, prices, trade, fuel use, and other time-series trends in comparison countries that are plausibly predictive of Iceland's pre-shock trajectory. Fields are excluded when coverage is incomplete for the required training window or when they are rounded estimates that are too noisy to improve prediction. For example, when the model training period runs from 2014 to 2019, a series that begins in 2015 cannot be used for that specification. Some survey-based series are rounded to the nearest 100 or 1,000, which can make them too noisy for monthly prediction.

Monthly data are important because Icelandic tourism is highly seasonal and because the relevant policy horizon after an airline bankruptcy is short. Monthly outcomes provide more opportunities to learn the pre-shock relationship between Iceland and comparison indicators, and they correspond to the time scale on which local officials, tourism firms, and labor agencies would need to respond. The post-shock outcomes are observed and estimated from April 2019 through December 2019 to avoid confounding effects from COVID-19. The shorter window limits what can be said about long-run recovery, but it protects the interpretation of the estimated effect.

Table 1: Observed Characteristics of Treated and Potential Control Countries

	Iceland	**Faroe Islands**	**Greenland**	**Norway**	**Denmark**	**Sweden**	**Ireland**	**UK**
Nordic Country	X	X	X	X	X	X	-	-
Distance to Iceland (mi.)	-	181	272	339	917	774	670	492
Seasonality (Difference Between Longest & Shortest Day)	17:02	14:37	17:04	12:56	10:31	12:32	9:30	8:48
Population 2018 (million)	0.35	0.05	0.06	5.31	5.79	10.18	4.87	66.46
Annual Tourists per Capita (2013)	2.78	2.41	1.22	0.94	4.72	1.14	1.79	0.52
Annual Tourists per Capita (2018)	7.05	3.66	1.52	1.07	5.32	0.73	2.24	0.61
5 Year Change in Tourism Per Capita	154%	52%	25%	14%	13%	-36%	26%	16%
Fish, Animals, and Animal Products as Export Product Share (2009)	38%	87%	56%	6%	11%	2%	4%	1%

Source: 1. Globefeed.com, 2. US Naval Observatory Astronomical Applications Department, 3. Nordicstatistics.org, 4. World Bank

One major advantage of the Bayesian structural time-series design is that it can use informative indicators even when data fields are not symmetric across countries. Iceland reports sector-specific tourism employment and wages, while some comparison countries report only aggregate labor or tourism indicators. This asymmetry is common for smaller destinations such as Greenland and the Faroe Islands. A strict matched-outcome design would discard many indicators because they are not measured identically across all countries. That restriction would be costly when the pool of plausible comparison destinations is already small. The preferred approach instead asks whether available indicators predict Iceland before the shock. Hotel nights in Greenland, passenger counts in the Faroe Islands, unemployment rates in Denmark, or fuel-use measures in Norway may be informative even when they are not exact replicas of the Icelandic outcome. These candidate predictors form a common pool, and spike-and-slab selection reduces the influence of weak predictors during estimation. The Supplemental Material reports the full data-availability matrix and charts showing how coverage differs across countries and over time.

2.3 Bayesian Structural Time-Series Design

The empirical objective is to estimate the effect of the WOW Air bankruptcy on Iceland's outcomes by comparing observed series with the counterfactual path that would have occurred in its absence. Let $Y_{I,t}$ be the observed Icelandic outcome in month t, and let $Y_{N,t}$ be the unobserved no-bankruptcy counterfactual. The period-specific causal effect α_t is:

$$\alpha_t = Y_{I,t} - Y_{N,t}.$$

Because $Y_{N,t}$ is not observed, it must be estimated from a model informed by Iceland's pre-bankruptcy trajectory, seasonality, and contemporaneous indicators from comparison economies. For monthly level outcomes, the average post-bankruptcy effect is reported as $\bar{\alpha}$ over April–December 2019. Relative effects divide the average difference by the average

counterfactual level. For flow variables such as passenger counts, the cumulative effect is also reported as $\sum_{t=n+3}^{m} \alpha_t$. A negative estimate should be interpreted as a shortfall relative to the synthetic counterfactual, not necessarily as an absolute decline in the observed series. If the observed outcome remained flat while the counterfactual continued to rise, the treatment effect would be negative because the outcome was lower than expected. This interpretation is the same as in a difference-in-differences design, but it is easier to see in this setting because the full time series makes the pre-shock growth path visible. The counterfactual $Y_{N,t}$ is predicted using the Bayesian structural time-series model presented by Brodersen et al. (2015) and implemented with Brodersen's `CausalImpact` package in R.[7]

Estimation is Bayesian. After fitting the model on the pre-intervention period, `CausalImpact` uses Markov chain Monte Carlo draws to simulate the posterior predictive distribution of the missing counterfactual path, $Y_{N,t}$, during the post-intervention period. Each draw implies a possible counterfactual, a pointwise treatment effect, and a cumulative treatment effect. The reported uncertainty intervals are therefore Bayesian credible intervals rather than frequentist uncertainty intervals. They describe the posterior probability that the effect lies within a given range, conditional on the model, priors, and observed data. The reported tail-area probabilities should be interpreted in the same posterior-predictive sense.

The preferred specification is:

$$Y_t = \mu_t + \tau_t + \sum_{j \in \mathcal{J}} x_{j,t}\beta_j + \varepsilon_t, \tag{1}$$

$$mu_{t+1} = \mu_t + \delta_t + \eta_{\mu,t}, \tag{2}$$

$$delta_{t+1} = \delta_t + \eta_{\delta,t}, \tag{3}$$

$$tau_{t+1} = -\sum_{k=1}^{S-1} \tau_{t-k} + \eta_{\tau,t}, \quad S = 12. \tag{4}$$

Equation (1) is the observation equation. Each Icelandic outcome, such as international passenger departures or tourism employment, is expressed as a local level, a seasonal compo-

[7] See the `CausalImpact` package documentation at `https://google.github.io/CausalImpact/`.

nent, selected contemporaneous predictors from comparison economies, and noise. Equation (2) defines the local level μ_t, representing Iceland's underlying trajectory. Equation (3) introduces a local slope δ_t, allowing that trajectory to change over time. Equation (4) captures monthly seasonality with $S = 12$, which is central in Icelandic tourism. The disturbances ε_t, $\eta_{\mu,t}$, $\eta_{\delta,t}$, and $\eta_{\tau,t}$ are treated as mutually independent Gaussian errors.

Four dates define each model: the start of the training period, the end of the training period, the beginning of the treatment window, and the end of the forecasting window. In notation, the pre-intervention training period is $t = 1, \ldots, n$, with $t = 1$ marking the training start and n corresponding to January 2019. February and March 2019 are excluded as a two-month transition period, $t = n + 1, n + 2$, because WOW Air reduced routes before its final bankruptcy in late March. The treatment window therefore begins in April 2019, $t = n + 3$, and the forecasting period continues through December 2019, $t = m$. This buffer avoids treating months that were already partially affected by the collapse as either clean pre-treatment observations or as the full shock. The preferred passenger model begins in 2014, when monthly coverage becomes richer and after Iceland's tourism sector had entered a high-growth period. The analysis keeps 2014 as the preferred start where possible for consistency, but some outcome-specific models begin later when the series of interest or substantively important covariates are not available earlier. Supplemental material Section S1 reports sensitivity to variation in training windows and provides additional model-calibration evidence.

Control selection is trained on the pre-bankruptcy period using spike-and-slab priors. The spike shrinks irrelevant coefficients toward zero, while the slab allows nonzero values for the most predictive controls. This is useful because the model can be offered a rich set of candidate covariates without forcing all of them into the counterfactual. Posterior inclusion probabilities summarize how often each predictor is selected across posterior draws. A high posterior inclusion probability means that a covariate is consistently useful for predicting Iceland's pre-shock path; the associated coefficient magnitude indicates the direction and

size of that predictive relationship conditional on inclusion. Supplemental Figure S3 reports posterior inclusion probabilities for the primary passenger model.

When interpreting the labor-market results, the estimates should be understood as deviations from a counterfactual path rather than direct counts of dismissed workers. Employment losses may combine actual job losses with foregone employment growth that would likely have occurred if the pre-shock trajectory had continued. This distinction matters for planning because layoffs, reduced hours, and missing seasonal hiring all affect income, firm capacity, tax receipts, and the demand for public support.

2.4 Identification and Interpretation

Identification requires the model to produce a plausible no-bankruptcy path after conditioning on Iceland's pre-shock trajectory, seasonality, and relationship with selected exogenous comparison indicators. One concern is that the bankruptcy could have coincided with broader shocks, such as fuel-cost changes, regional travel demand shifts, or financial pressure affecting low-cost carriers more generally. The use of contemporaneous comparison indicators helps mitigate this concern. Shocks that also affect nearby economies should be partly reflected in the predictor series rather than attributed entirely to Iceland. A second concern is spillover. WOW Air served routes connected to other countries, including Denmark, and a loss of Iceland-connected traffic could in principle affect comparison economies. This concern would most likely bias the estimates toward zero if it reduced passenger or tourism indicators in comparison countries, because the predicted Icelandic counterfactual would also be pulled downward. **Additionally, because the primary passenger outcome measures total departures rather than WOW Air passengers, any substitution to other carriers is already incorporated in the observed series and would attenuate, rather than generate, the estimated shortfall.** The placebo checks in the Supplemental Material provide additional support by showing that the same bankruptcy timing does not generate comparable effects in untreated passenger markets or unrelated Icelandic outcomes.

The passenger channel is central for interpreting the labor estimates. Without a clear decline in passenger departures, labor-market effects would be difficult to attribute to the airline collapse beyond the airline's own workers. **The empirical sequence is to first establish that passenger flows declined following the loss of airline service,** then to estimate whether labor outcomes moved below their counterfactual paths, and finally to evaluate whether placebo and robustness tests support that interpretation. **Added nationality-level disaggregation shows that the aggregate decline was not driven solely by changes in resident travel, and hints at which markets were most affected.**

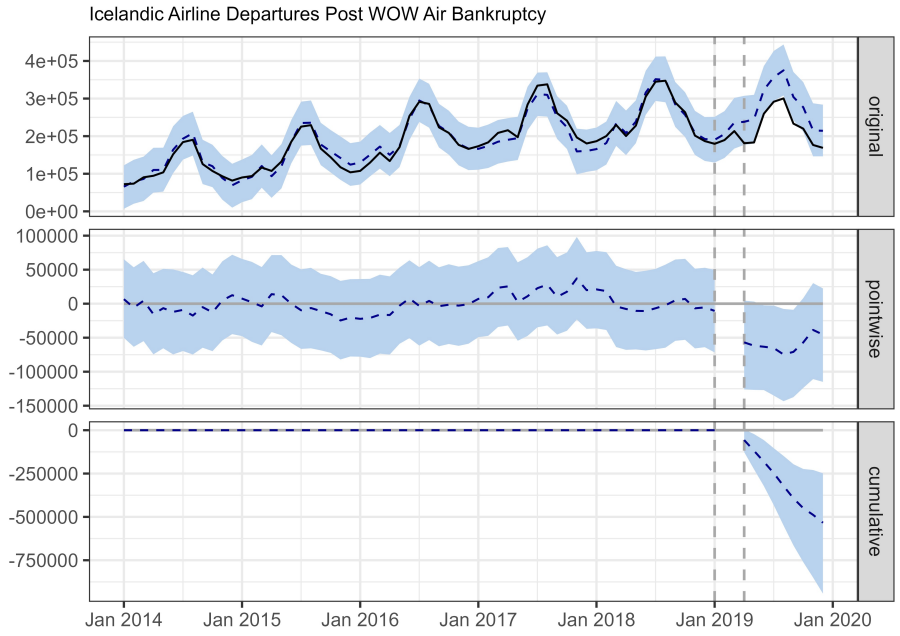
The estimates do not imply that every 2019 labor-market change was caused by WOW Air. The large growth rate of Icelandic tourism had already begun slowing prior to the bankruptcy, and the Supplemental Material placebo tests show some evidence of a smaller pre-2019 slowdown. The preferred model incorporates that pre-2019 information rather than comparing 2019 to an indefinitely rising trend. The effects should be read as short-run deviations from the expected path conditional on the information available before the final bankruptcy. This same rationale explains why a standard difference-in-differences design is less suitable for this setting. Iceland's tourism boom was unusually rapid relative to nearby countries, making parallel trends unlikely. A two-way fixed-effects model would require the comparison economies to represent Iceland's untreated trend after fixed effects and seasonality. The Supplemental Material shows that those estimates are unstable across pre-periods and produce implausible placebo effects. The Bayesian structural time-series design is better matched to a rapidly growing destination because the counterfactual can depend on Iceland's own time-series structure and selected predictors from comparison economies.

3 Results

3.1 Passenger Effects

Figure 1 visualizes the shock to passenger flows. The figure compares observed monthly international departures with the model's no-bankruptcy counterfactual. In the top panel, the solid line reports observed departures, the dashed line is the expected path absent the bankruptcy, and the shaded interval shows posterior uncertainty. The second panel reports monthly differences between observed and counterfactual departures, and the bottom panel reports the cumulative post-bankruptcy effect. From April through December 2019, total international passenger departures were about 534,000 below the counterfactual, a 20.6% shortfall relative to expected departures.

Figure 1: Observed and Counterfactual International Passenger Departures



Note: Departures are measured as the number of persons passing through international security in Iceland. The solid line shows observed values, the dashed line shows the Bayesian structural time-series counterfactual, and shaded areas indicate posterior uncertainty.

Table 2 reports the estimates shown in Figure 1. Each row compares an observed Icelandic

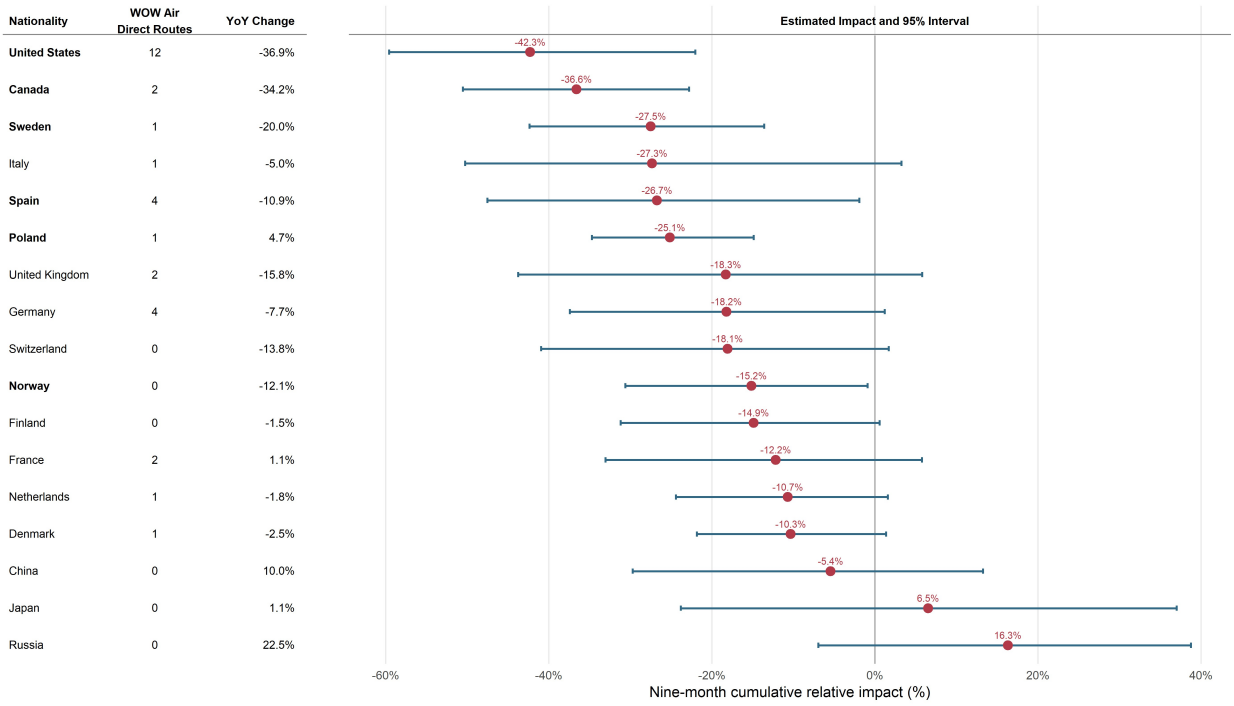
passenger series with its Bayesian structural time-series counterfactual for April–December 2019. **The first three rows separately estimate total passengers, Icelandic passengers, and foreign passengers.** The relative-impact column reports the difference between observed and counterfactual departures as a percentage of the counterfactual. The cumulative-impact column reports the total April–December passenger difference in levels. The credible-interval columns report posterior uncertainty for the cumulative effect, the p-value reports the fitted model's posterior tail-area probability, and RMSE reports pre-period prediction error.

Table 2: Passenger Effects of the WOW Air Bankruptcy, by Passenger Nationality

Passenger Series	Relative Impact (%)	9-Month Cumulative Impact	95% CI Upper	95% CI Lower	p-value	RMSE (pre-period)
Total Passengers	-20.63***	-533,643	-251,386	-942,546	0.0002	13,530
Icelandic Passengers	-16.70***	-100,771	-23,111	-180,957	0.0010	2,905
Foreign Passengers	-22.75***	-461,168	-211,729	-831,090	0.0002	11,895
United States	-42.29***	-272,408	-141,886	-383,908	0.0002	6,063
Canada	-36.59***	-32,481	-20,250	-44,873	0.0002	1,153
Sweden	-27.51***	-12,861	-6,352	-19,808	0.0008	499
Italy	-27.34**	-13,537	1,601	-24,885	0.0410	442
Spain	-26.75**	-17,775	-1,284	-31,582	0.0153	627
Poland	-25.14***	-25,728	-15,218	-35,515	0.0002	568
United Kingdom	-18.29*	-33,566	10,633	-80,275	0.0696	3,290
Germany	-18.21**	-24,022	1,594	-49,395	0.0364	1,427
Switzerland	-18.09**	-4,816	448	-10,903	0.0384	287
Russia	16.31*	2,115	5,027	-901	0.0778	171
Norway	-15.16**	-5,735	-344	-11,590	0.0200	306
Finland	-14.88**	-3,100	116	-6,500	0.0297	260
France	-12.17	-10,769	5,094	-29,234	0.1000	851
Netherlands	-10.70**	-4,186	617	-9,561	0.0475	419
Denmark	-10.33**	-4,614	610	-9,752	0.0408	340
Japan	6.51	765	4,346	-2,797	0.3503	268
China	-5.45	-4,453	10,830	-24,292	0.3040	1,027

Note: Estimates compare April–December 2019 observed passenger departures with the no-bankruptcy counterfactual. The 95% intervals and nine-month impacts are in passenger counts. Stars denote statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 2: Passenger Impacts by Nationality and WOW Air Direct Routes



Note: Points show nine-month cumulative relative passenger impacts with 95% intervals. The left columns report passenger nationality, WOW Air direct routes in 2018, and observed year-over-year change.
Source: WOW air hf. (2018), Investor Presentation: Senior Secured Bond Issue, August 2018, Slide 12. Company data.

Foreign passengers accounted for most of the absolute shortfall: about 461,000 fewer foreign passenger departures, or 22% below expected, compared with about 101,000 fewer Icelandic passenger departures, or 16% below expected. Icelandic residents also lost travel options when WOW Air exited, but resident travel from Iceland is not the main channel for local tourism employment impacts.

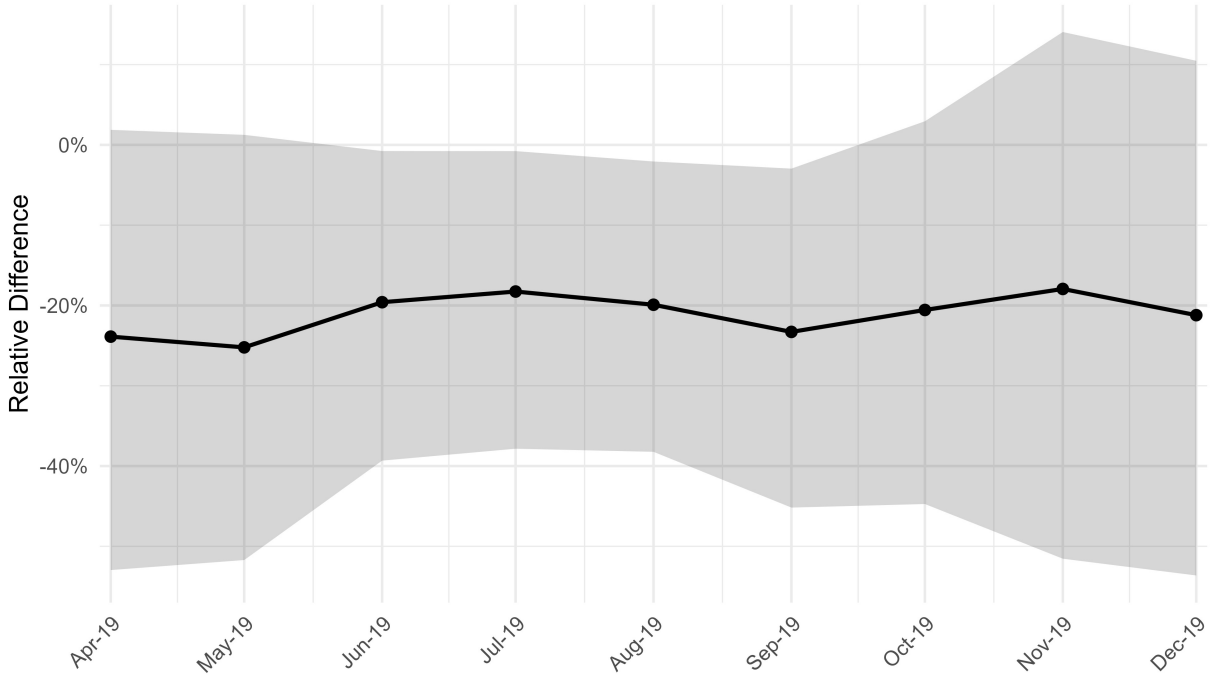
The nationality-level estimates reveal substantial heterogeneity in the passenger shortfall across markets. However, Icelandic passenger data identify nationality rather than trip origin, so nationality is not equivalent to source market and does not establish that a passenger traveled on a particular WOW Air route. The number of direct routes also measures only one dimension of WOW Air's presence and does not capture flight frequency, capacity, or fares. Nonetheless, Figure 2 shows stark differences in passenger outcomes across nationalities and suggests that some of the largest declines occurred in markets with substantial WOW Air presence.

The largest declines appear in North America. From April through December 2019, passenger departures among U.S. and Canadian nationals were 36.9% and 34.2% lower year over year, respectively. The counterfactual estimates indicate even larger shortfalls relative to the no-bankruptcy path, at 42.3% for the United States and 36.6% for Canada. WOW Air operated 14 direct routes to these two markets in 2018, including 12 to the United States.

The results also contrast countries where WOW Air had a large presence with those where it operated no direct routes. This is especially notable for more geographically distant markets such as Japan and Russia, where the estimated effects are positive but imprecise and indistinguishable from zero. Norway is a notable exception, with a statistically significant decline despite no direct WOW Air route. However, European markets are also more difficult to interpret

because nationality does not necessarily correspond to airport of origin. Travelers may readily use airports in neighboring countries, so the route comparison should not be interpreted as a formal relationship between WOW Air service and country–specific effects. Nonetheless, these patterns therefore suggest that the bankruptcy affected not only the total number of passengers reaching Iceland, but also the composition of those passenger flows across nationalities.

Figure 3: Monthly Passenger Departures Relative to the Counterfactual



Metric	Apr-19	May-19	Jun-19	Jul-19	Aug-19	Sep-19	Oct-19	Nov-19	Dec-19
Actual	181.2k	183.3k	259.7k	291.6k	300.4k	233.4k	219.9k	176.5k	168.8k
Control	238.1k	245.1k	323.0k	356.8k	375.0k	304.2k	276.9k	215.1k	214.3k
Difference	-56.9k	-61.8k	-63.3k	-65.2k	-74.7k	-70.9k	-56.9k	-38.6k	-45.4k

Note: The solid line shows the estimated monthly percentage difference between observed and counterfactual departures. Shaded areas represent 95% credible intervals. The table embedded in the figure reports actual, predicted, and difference values for each post-shock month.

Although not formally testable under this methodology, the passenger estimates also show limited attenuation during the nine-month post-bankruptcy window. If switching to other carriers had offset the loss of WOW Air passen-

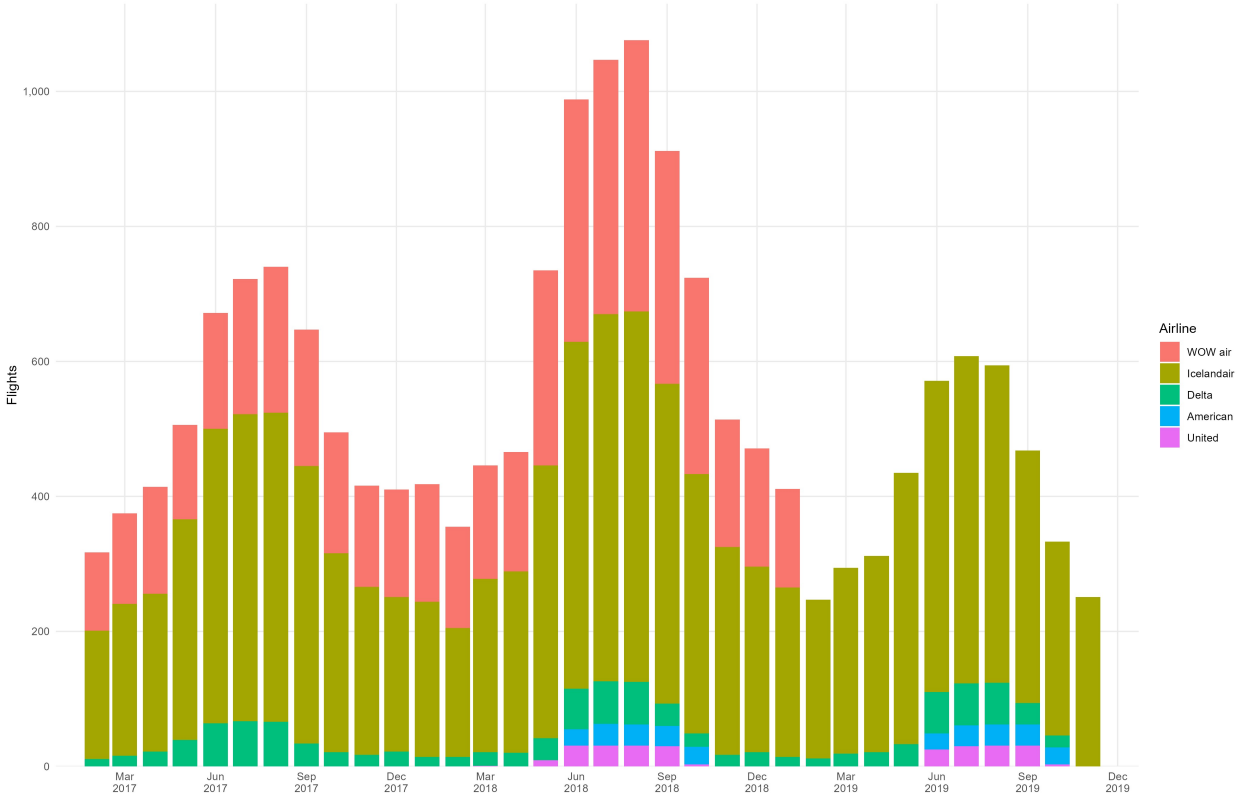
gers, the observed series would be expected to move closer to the counterfactual. Instead, the gap between expected and observed departures persisted across the post-shock window. Monthly departures remained close to 20% below expected levels in most months, with no clear return to the counterfactual path before the end of 2019, as shown in *Figure 3*. The persistent shortfall appears to suggest that WOW Air's passenger flows were not quickly replaced after its exit.

Additional descriptive evidence from the U.S. market supports this interpretation. *Figure 4* shows monthly U.S.–Iceland flights by carrier from 2017 through 2019. WOW Air's flights disappear following the bankruptcy, while the remaining carriers continued operating without appearing to increase service enough to restore the total number of flights observed in 2018.

3.2 Labor-Market Effects

Table 3 reports whether the passenger shock reached employment, unemployment, and wages. Each row compares an observed Icelandic labor-market or wage outcome with its Bayesian structural time-series counterfactual for April–December 2019. The table is organized in three panels: national labor indicators, employment by sector, and taxable wages by sector. The nine-month relative-impact column combines the point estimate, significance stars, and 95% interval. The next column reports the observed year-over-year change for April–December 2019 relative to the same months of 2018. Average monthly impact is the observed-minus-counterfactual difference in original units: workers for employment and unemployment, percentage points for the unemployment rate, and ISK millions for wages. The following columns give monthly average predicted and observed levels, followed by the p-value and pre-period RMSE.

Figure 4: Monthly Flights from U.S. to Iceland by Airline, 2017–2019



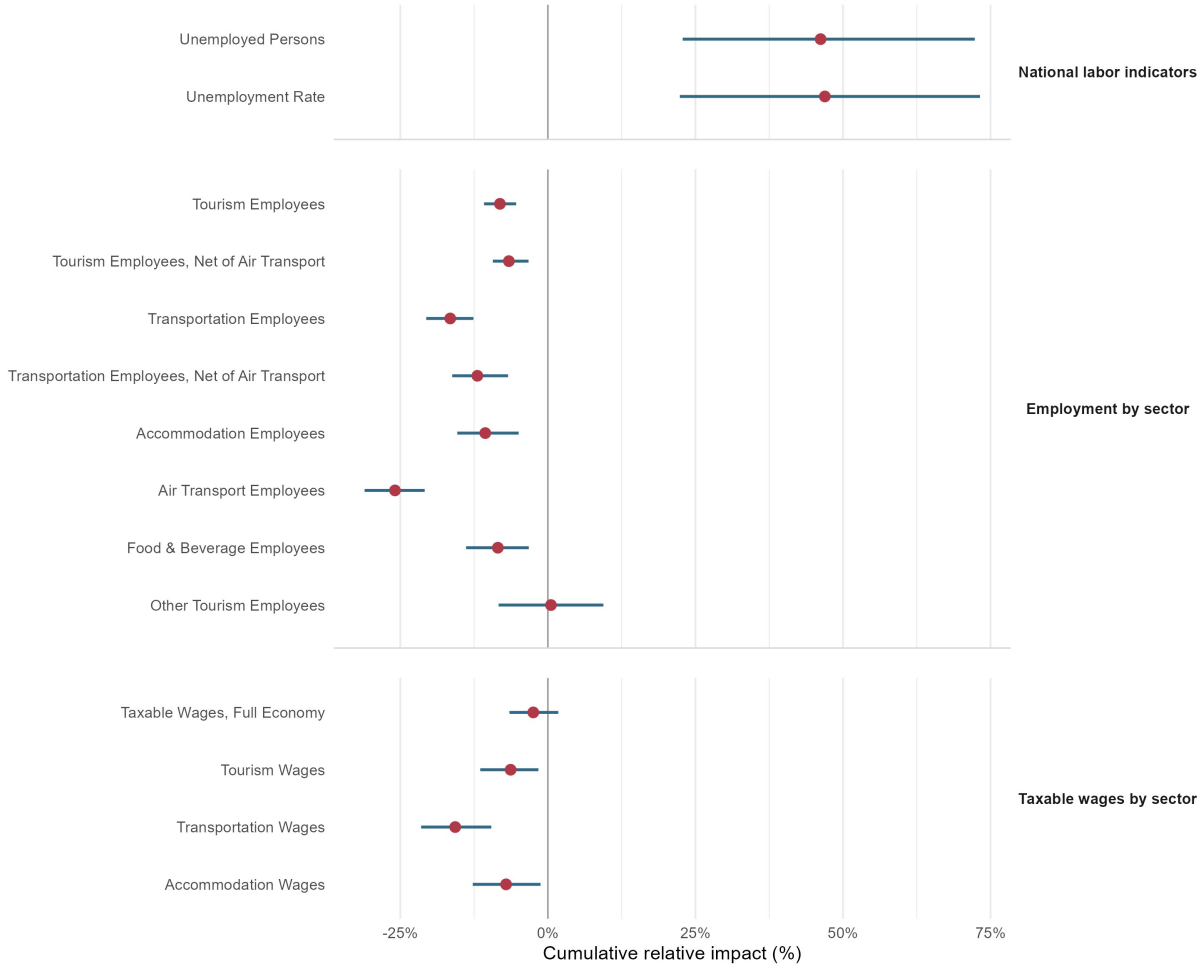
Note: Stacked bars show monthly U.S.–Iceland flight counts for the five airlines shown. Source: U.S. Department of Transportation, Bureau of Transportation Statistics, T-100 International Segment (All Carriers).

Table 3: Labor-Market and Wage Effects of the WOW Air Bankruptcy

Outcome	9-Month Relative Impact (%) and 95% CI	YoY Actual Change (%)	Avg. Monthly Impact	Monthly Avg. Predicted	Monthly Avg. Observed	p-value	RMSE (pre-period)
National labor indicators							
Unemployed Persons	46.21*** [22.8, 72.3]	32.62	2,613	5,654	8,267	0.0006	1,263
Unemployment Rate	46.93*** [22.4, 73.2]	30.82	1.30	2.76	4.06	0.0014	0.66
Employment by sector							
Tourism Employees	-8.12*** [-10.8, -5.4]	-3.93	-6,010	73,995	67,985	0.0002	739
Tourism Employees, Net of Air Transport	-6.61*** [-9.3, -3.3]	-2.49	-4,498	68,072	63,574	0.0010	765
Transportation Employees	-16.54*** [-20.6, -12.6]	-12.03	-2,653	16,034	13,382	0.0002	200
Transportation Employees, Net of Air Transport	-11.94*** [-16.2, -6.7]	-6.99	-1,217	10,187	8,971	0.0002	162
Accommodation Employees	-10.61*** [-15.3, -4.9]	-4.09	-2,052	19,345	17,293	0.0010	567
Air Transport Employees	-25.90*** [-31.0, -20.9]	-20.76	-1,542	5,953	4,411	0.0002	116
Food & Beverage Employees	-8.47*** [-13.9, -3.2]	-6.55	-939	11,084	10,144	0.0002	167
Other Tourism Employees	0.53 [-8.3, 9.4]	23.09	169	32,143	32,311	0.4539	36
Taxable wages by sector							
Taxable Wages, Full Economy	-2.47 [-6.5, 1.8]	3.20	-2,916	117,902	114,986	0.1175	1,233
Tourism Wages	-6.31*** [-11.5, -1.6]	0.03	-2,065	32,710	30,644	0.0042	390
Transportation Wages	-15.70*** [-21.5, -9.6]	-8.48	-1,564	9,960	8,397	0.0002	159
Accommodation Wages	-7.08*** [-12.7, -1.2]	1.57	-448	6,319	5,871	0.0075	131

Note: Relative impact and its 95% interval compare cumulative observed and counterfactual outcomes over April–December 2019. YoY actual change compares the observed nine-month average with the same months of 2018. Monthly quantities and RMSE are measured in workers for employment and unemployment, percentage points for the unemployment rate, and ISK millions for wages. Stars denote statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 5: Nine-Month Cumulative Relative Labor Impacts



Note: Points show April–December 2019 cumulative relative impacts compared with the no-bankruptcy counterfactual. Horizontal bars show 95% intervals.

The largest employment effects appear in sectors closest to the **disruption** shown in Figure 1. Air-transport employment predicts an average of 1,542 fewer employees than the counterfactual, higher than the approximately 1,000 employees that would be immediately laid off as a result of WOW Air's closure.[8] Tourism employment net of air transport was 6.6% below expected levels, or about 4,498 fewer workers per month, and transportation employment net of air transport was 11.9% below expected levels, or about 1,217 fewer workers per month. The labor-market shortfalls therefore extended beyond the airline itself. Transportation services, accommodation, food and beverage, tour services, and other visitor-facing positions depend on the same passenger flows. Supplemental Table S7 reports supporting indicators showing reduced rental-car and traffic activity after the bankruptcy. The results do not imply that every sector moved equally but they show that the impacts of the shock extended beyond the airline itself.

National unemployment increased as well, both relative to the counterfactual and year over year. The number of unemployed persons was 46.2% higher than expected, or about 2,613 additional unemployed persons per month. The unemployment rate was also nearly 50% above the counterfactual. The percentage effect is large partly because expected unemployment was low, but the level effect is meaningful in a national labor force of roughly 200,000 workers. Supplemental Figure S6 shows the monthly labor-market deviations from the counterfactual. Although the monthly estimates are noisier than the passenger estimates, the increase in unemployment is visible across the post-bankruptcy period. **This distinction is especially important for employment outcomes because the model compares observed employment with the path expected in the absence of the bankruptcy. The resulting gap may reflect both jobs that were lost and employment growth that did not occur.**

Unemployment did not increase one-for-one with tourism employment shortfalls. Employment adjustment likely occurred through layoffs, reduced hiring, movement into other

[8]See CNBC, "Icelandic airline WOW Air collapses and cancels all flights," `https://www.cnbc.com/2019/03/28/icelandic-airline-wow-air-collapses-and-cancels-all-flights.html`.

sectors, reduced hours, and exits from measured employment. Wage totals provide a complementary measure because they can respond when hours fall before head counts change. **As with employment, wage estimates are measured relative to the counterfactual path rather than as simple year-over-year losses.** The strongest wage effect appears in transportation, the sector closest to the airline collapse, while the tourism-wage effect shows that the consequences were not limited to head counts. Transportation taxable wages were 15.7% below the counterfactual, equal to an average monthly shortfall of 1,564 million ISK. Over the nine-month post-bankruptcy period, that implies roughly 14.1 billion ISK in missing taxable wages, or about $114 million in 2019 USD.[9] Tourism taxable wages were 6.3% below the counterfactual, equal to an average monthly shortfall of 2,065 million ISK. Over the same period, that implies roughly 18.6 billion ISK in missing taxable wages, or about $151 million in 2019 USD. Economy-wide taxable wages were also below expected levels, and the point estimate is larger in absolute terms, but the estimate is noisy and not statistically significant. The table therefore does not show an economy-wide wage collapse. It does, however, show large sectoral wage losses in the parts of the economy closest to the passenger shock.

These wage and unemployment results also give the labor-market shock a public-finance interpretation. The missing taxable wages are not only private income losses for workers and firms. They also represent income that would have generated tax revenue, consumption, and economic activity. **At the same time, registered unemployment increased by about 2,600 persons per month relative to the counterfactual, potentially increasing demands on unemployment support at the same time that taxable earnings were falling. This is not a failure of social insurance, but rather one of the fiscal consequences of a labor-market shock. An airline bankruptcy could affect public finances through both lower tax-generating earnings and greater**

[9]Dollar conversions use an approximate 2019 exchange rate of 0.00810 USD per ISK. The conversion is intended only to make the magnitude more interpretable. The empirical estimates are reported in Icelandic krona.

use of labor-market support programs. Supplemental Figure S6 illustrates the increase in unemployment during the post-bankruptcy period. The estimates may also understate the full adjustment problem because the analysis cannot observe whether effects would have persisted beyond 2019 in the absence of the global disruption caused by COVID-19.

3.3 Robustness Overview

The Supplemental Material reports three sets of validity checks, focusing on the primary passenger outcome because it is the primary measure of the passenger-flow response. Placebo tests guard against false positives by applying the same treatment timing to outcomes or countries that should not have been affected. The bankruptcy timing does not produce a statistically significant effect for Icelandic fish landings or for passenger outcomes in Greenland, Denmark, or Norway. None of these should be measurably affected by WOW Air's bankruptcy. A 2018 pre-period placebo applies the same treatment structure to Iceland one year earlier and produces a smaller, weaker negative effect, suggesting that tourism was cooling before the bankruptcy. Because the preferred 2019 model is trained through January 2019, it incorporates the pre-event slowdown. This does not prove that the entire effect is attributable only to WOW Air, but it reduces the concern that the estimate is simply extrapolating a pre-2019 boom indefinitely.

Sensitivity checks vary the control-country set and covariate grouping. The passenger decline remains large and statistically significant when individual control countries are omitted and when different pools of covariates are used. These checks are important because the comparison set is small and the available indicators differ across countries. The main conclusion does not depend on a single comparison economy or one narrow predictor group.

The Supplemental Material also compares the preferred model with simpler alternatives. A model without time-series components produces a larger passenger-loss estimate but worse pre-shock fit. This suggests that the preferred model is more conservative and better aligned with a highly seasonal tourism series. A two-way fixed-effects comparison is less stable:

changing the pre-period can switch estimates from negative to positive and can also produce effects for control countries. These checks support the Bayesian structural time-series approach in a rapidly growing destination where parallel trends are unlikely.

3.4 Pre-Shock Exposure

Iceland's pre-shock specialization may help explain why the labor-market response was so large. Table 4 reports annual averages for 2010, 2018, and 2019. Each row reports a different measure of tourism exposure, including foreign visitors per worker and employment shares in tourism-facing sectors. Monthly foreign visitors per worker rose from 0.24 in 2010 to 0.99 in 2018. Visitors per worker is a simple ratio, but it shows how quickly visitor volume grew relative to the domestic labor force. Tourism employees increased from 7.9% of all employees in 2010 to 14.8% in 2018. Air transport nearly doubled as a share of total employment, and tour services more than tripled. A sudden decline in visitors therefore changes the operating environment for firms that staffed for continued growth. Supplemental Figure S7 complements the annual tourism exposure table by showing the monthly post-bankruptcy deviations for selected labor outcomes.

Table 4: Tourism Employment Concentration in 2010, 2018, and 2019

Indicator	2010	2018	2019	% Change, 2010-2018	% Change, 2018-2019
Tourists per Worker					
Monthly Foreign Visitors Per Worker	**0.24**	**0.99**	**0.85**	**314.61%**	**-14.10%**
Employment Breakdown By Tourism Sector					
Tourism Employees as % of All Employees	**7.88%**	**14.84%**	**13.96%**	**88.41%**	**-5.97%**
Transportation Employees as % of All Employees	5.18%	7.56%	6.85%	45.81%	-9.40%
Air Transport Employees as % of All Employees	1.46%	2.72%	2.28%	86.67%	-16.17%
Accommodation Employees as % of All Employees	5.42%	8.98%	8.66%	65.88%	-3.54%
Food and Bev Employees as % of All Employees	3.91%	5.48%	5.16%	40.41%	-5.84%
Tour Services Employees as % of All Employees	0.60%	1.98%	1.97%	232.45%	-0.49%
Other Tourism Employees as % of All Employees	0.63%	1.45%	1.35%	130.35%	-6.64%

Note: Values are annual averages calculated from monthly data. The first three year columns show the level of each exposure measure. The final two columns show relative changes before the bankruptcy period (2010–2018) and across the post-bankruptcy period (2018–2019).

Tourism employment quickly transformed post-bankruptcy, falling from 14.8% of all employment in 2018 to 14.0% in 2019. Air-transport employment fell from 2.7% to 2.3%. Iceland entered 2019 with a labor market increasingly specialized in tourism-facing work, and the bankruptcy reduced the passenger flows supporting that specialization.

Despite Iceland's growth and specialization, the case may still generalize to destinations with concentrated air service and tourism-facing labor exposure. Iceland is a small island economy, but it is still more diversified than many island tourism destinations. Several Caribbean and small-island economies have even larger tourism shares of GDP, employment, or export earnings, meaning that a similar shock could have larger aggregate consequences in more specialized settings. In 2019, travel and tourism accounted for more than half of total employment in each of the ten most tourism-dependent destinations, ranging from 51.3% in Anguilla to 90.7% in Antigua and Barbuda.[10] The Iceland case is therefore not an extreme upper bound. **Rather, it shows how dependence on transportation can translate into broader labor-market exposure, even in a relatively high-income and diversified island economy.**

[10]See World Economic Forum, "These are the 10 places that rely most on tourism for jobs," `https://www.weforum.org/stories/2020/08/destinations-rely-most-on-tourism-travel/`. The figure is attributed there to WTTC Economic Impact Research; see World Travel & Tourism Council, "Travel & Tourism Economic Impact Research," `https://wttc.org/Research/Economic-Impact`.

4 Limitations

Iceland's passenger flows were still increasing before the bankruptcy, but in 2018, growth had slowed. Total passenger departures from March through December grew by approximately 19% in 2017 and 6% in 2018, before falling by 14% in 2019.[11] Some portion of the 2019 passenger decline could therefore reflect a broader slowing of Icelandic tourism growth rather than WOW Air alone. This concern is highlighted by the placebo test presented in Supplemental Table S2, which applies the same treatment setup one year earlier to test whether the model detects a similar effect when no bankruptcy occurred. It produces a 7.5% negative estimate and provides weak evidence against the null ($p = 0.0742$). This may conflate slowed growth with the bankruptcy. However, the preferred 2019 model is trained through January 2019 and therefore directly incorporates the slower 2018 trajectory when constructing the counterfactual. Descriptive evidence from the U.S. market shows WOW Air's flights disappearing without an apparent offsetting increase among the other carriers shown in Figure 4. Together, the incorporation of the 2018 slowdown, the robustness checks, and the U.S. flight evidence reduce the concern that the estimated passenger effect simply reflects an ongoing pre-bankruptcy trend. Nevertheless, some upward bias in the estimated effect remains possible.

[11] Year-over-year calculations compare March–December total passenger departures in Statistics Iceland, SAM02001.

5 Discussion

WOW Air was a low-cost carrier whose growth coincided with, and likely helped enable, Iceland's tourism boom. Its collapse removed routes and low-cost travel options and abruptly interrupted that growth. The reduction in foreign passengers was accompanied by measurable short-run changes in tourism-facing labor markets. International passenger departures were about 20.6% below the counterfactual from April through December 2019, with foreign travelers accounting for most of the shortfall. Over the same period, tourism employment was more than 8% below its projected path, and tourism employment net of air transport was still 6.6% lower than expected. Registered unemployment also rose sharply. These results show that a tourism-dependent economy can face substantial labor-market risk when the transportation services supporting visitor flows are concentrated or difficult to replace.

The resulting labor-market outcomes also show that the shock was not self-contained. The most immediate and direct losses are understandably connected to the failed airline and the air-transport sector. However, effects were not limited to the airline itself. Transportation, broader tourism employment, sectoral wages, and registered unemployment also deviated substantially relative to the counterfactual. For a destination, the relevant issue is not only how many airline employees lose their jobs. It is also whether hotels, tour operators, restaurants, ground transportation firms, and other visitor-facing businesses face fewer customers, reduced hours, slower hiring, and lower wage payments. The results indicate this is a likely outcome.

The policy implication is therefore not airline rescue, but exposure planning. The results show that the negative tourism shock had both private labor-market consequences and a public-finance dimension. Taxable wages fell in affected sectors while registered unemployment increased, a combination that creates a political and financial opening for soft-landing policies aimed at affected workers, firms, and sectors. Destination agencies should understand which systems of the visitor economy depend on concentrated connections. That includes the share of service provided by a major carrier, the origin markets served by few

airlines, the role of low-cost service, the seasonality of routes, whether alternative service is available, and the sectors most tied to the affected visitor segments. If tourism grows faster than other industries, it is increasing reliance on these systems, and it is important to plan accordingly.

Additionally, we observe that some routes and markets, particularly those served by low-cost carriers, may not be quickly replaced by other airlines. This matters not only for the total number of visitors, but also for who those visitors are. When departures among U.S. nationals fall by more than 40% relative to expectations, the nationality composition of passenger flows changes substantially. Businesses may consequently face changes in customer relationships, spending patterns, exchange-rate exposure, and the returns to prior market-specific advertising and promotion. Hotels, tour operators, and destination marketers that had built products, partnerships, or campaigns around rapidly growing North American demand may therefore be affected differently from firms oriented toward European or other markets. Route losses can thus reshape both the scale and the composition of tourism demand, creating adjustment costs even when some passenger traffic is eventually replaced.

On a more optimistic note, the findings also show why the WOW Air case should not be viewed solely as a story of fragility. WOW Air appears to have contributed meaningfully to Iceland's pre-2019 tourism growth by expanding low-fare access, connecting markets, and helping sustain a larger visitor economy. That growth created employment and supported a more specialized tourism sector. The results are consistent with incomplete replacement of WOW Air's service by other carriers, with the U.S. market providing additional descriptive evidence that the lost flights were not quickly offset. The policy lesson is therefore not that low-cost carrier growth is undesirable. The more useful conclusion is that carrier-led growth should be paired with planning for what happens if those new connections are severed. More broadly, destinations need to know when visitor flows depend on a small number of carriers,

routes, or low-cost service.

By comparing observed outcomes with a synthetic no-bankruptcy path, this study shows how the sudden loss of air service can translate into substantial shortfalls in visitor flows and how those shortfalls may be accompanied by broader labor-market effects. For small, island, and peripheral destinations in particular, tourism planning should consider not only how to attract visitors, but also how resilient the transportation relationships supporting those visitors are to disruption.

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The data are drawn from public statistical agencies and other public sources described in the manuscript. Replication materials are available from the author upon reasonable request.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this manuscript, the author used OpenAI's ChatGPT for editorial assistance, including language revision, organization, and formatting suggestions. The author reviewed and completed the analysis, interpretation, and final manuscript content and takes full responsibility for the work.

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Supplemental Material

S1 Data Availability and Model Calibration

This section provides additional detail on data availability, covariate selection, and pre-period calibration. The method used in the main analysis was selected to address the need for monthly data and the uneven availability of comparable indicators across small North Atlantic and Nordic economies. The tables and figures below show why a flexible Bayesian structural time-series design is useful in this setting and how the preferred model is calibrated.

Table S1 reports the availability of monthly indicators by topic area and comparison economy. The first column lists the measure. The remaining columns show whether the measure is available for Iceland, the Faroe Islands, Greenland, Denmark, and Norway, and whether it is available as an aggregate or with additional detail. The table illustrates why conventional synthetic control methods are difficult in this setting. In the standard synthetic-control framework, the treated outcome is approximated by a weighted combination of the same outcome observed in control units (Abadie, Diamond and Hainmueller, 2010). That structure would require, for example, air-transport employment in Iceland to be predicted using comparable air-transport employment series in other countries. Table S1 shows why that requirement is restrictive. Iceland reports detailed sectoral tourism and labor outcomes, while some comparison economies report only aggregate indicators. The Bayesian structural time-series model addresses this limitation by allowing all available indicators to contribute to prediction when they improve pre-shock fit.

Figure S1 shows the share of potential covariates in the compiled dataset that are observed without missing values over time. Data coverage improves sharply by 2014. As a result, the preferred model training period begins in 2014 because this start date balances the need for enough pre-shock time-series information with the need for a rich set of monthly comparison indicators.

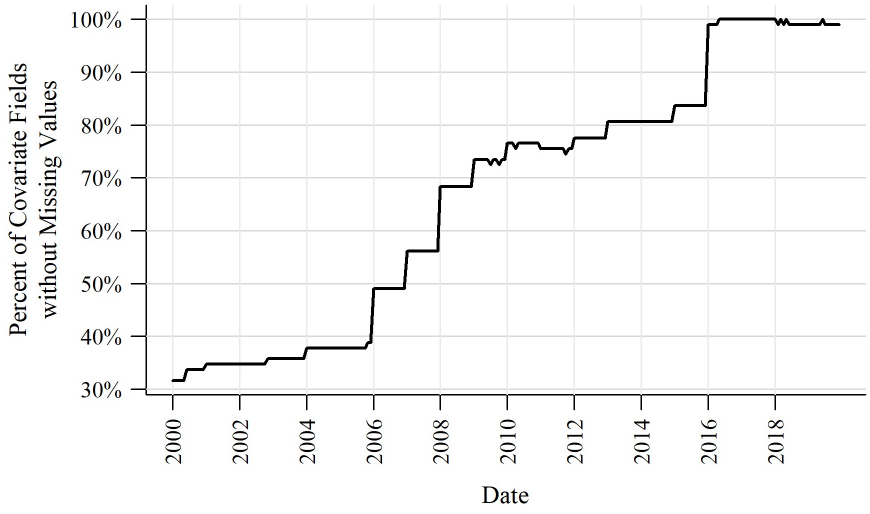
Figure S2 reports cumulative passenger effects across alternative pre-period start dates.

Table S1: Varying Data & Information Availability

Measure	Icelandic	Faroe Islands	Greenland	Denmark	Norway
Tourism & Transportation					
Airport Passengers	Arrivals, Departures by Citizenship	Arrivals	International, Total	Passengers	Passengers
Cruise Passengers	Total	-	Total	-	-
Hotel Occupancy Rate	By Bed, Room, by Type	-	-	Total	Total
Hotel Check-ins	Total, Foreign, Domestic, by Type	Total (Room Nights)	Total (Guests)	-	Total
Overnight Stays	Total, Foreign, Domestic, by Type	Total	Total	Total, Foreign	Guest Nights (Total, Foreign)
Hotel Rooms Available	-	-	-	Total	-
Rental Car Registrations	Total	-	-	-	-
Labor					
Unemployment Rate	Total, By Age	Total, By Age	Total	Total, By Age Group	Total, By Age
Unemployed Persons	Total, By Age	By Age	Total, Job Seekers Total, Under 25	Total, By Age Group	Total, By Age
Employed Persons	Total, By Age, By Sector	Total, By General Industry	Total, By Sector	Total, By Sector	Total Industry, By Sector, By Age
Wages	Total By Sector	Total, By Sector	-	-	Total Industry, By Sector
Hours Worked	Total, By Age	-	-	-	-
Insolvencies	Total, By Sector	-	-	Total, By Sector	Total, By Sector
Business Openings	-	-	-	Total, By Sector	-
National Accounting					
CPI	Total	-	-	Total	Total
Exports	Total	Total	Total	Total	Total
Imports	Total	Total	Total	Total	Total
Unemployment Rate	Total	Total	Total	Total	Total
Employment Counts	Total	Total	Industry Total	Industry Total	Industry Total
Insolvencies	Total	-	-	Total, By Sector	Total, By Sector
Business Creation	-	-	-	Total, By Sector	-
Other					
Local Traffic Count	Total, by Region	-	-	-	-
Fuel Use	Total, Paid by Foreign, Domestic	-	-	Total	Total
Fish Landings	Weight	By Weight / Currency	By Weight / Currency	-	Weight

Note: Reported measures by country showing variation in availability and level of detail across topic areas.

Figure S1: Monthly Covariate Series with Complete Data Coverage

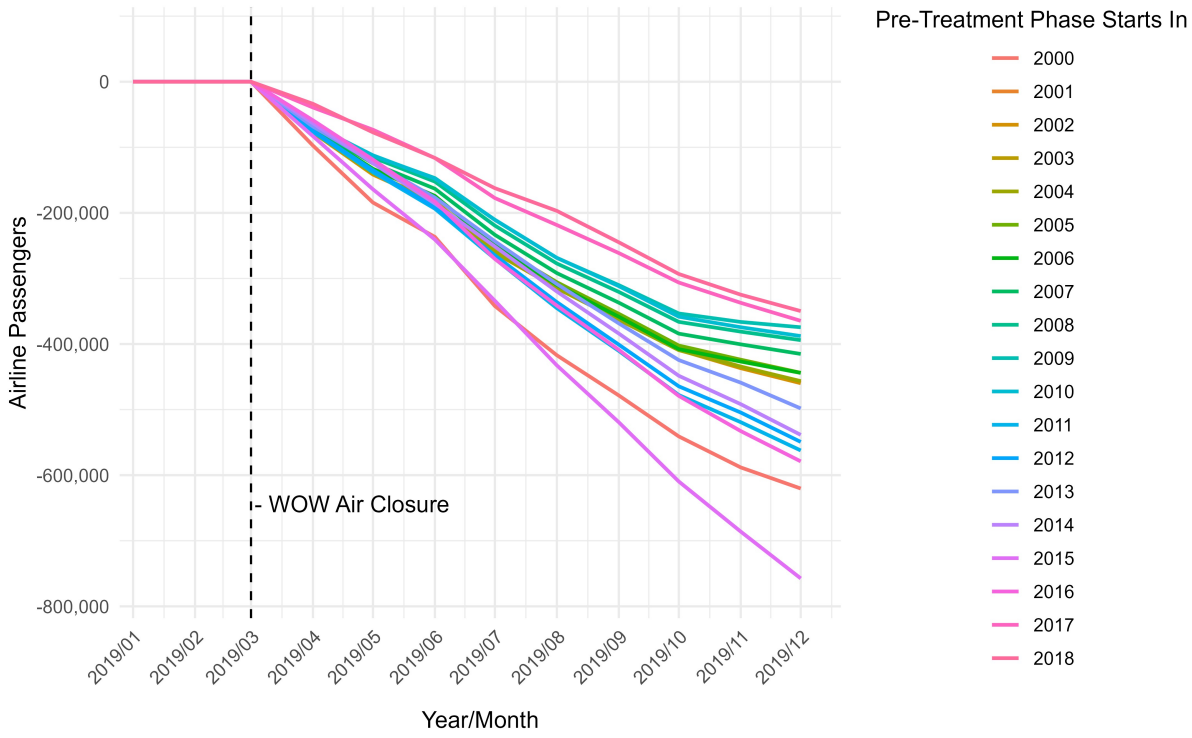


Note: The figure reports the percentage of candidate covariate fields with non-missing monthly values by date. Higher coverage allows richer counterfactual prediction.

The calibration exercise supports using 2014 as the main training start. Beginning much earlier provides more pre-period observations but substantially reduces available covariate data and reaches further back into a different period of Icelandic tourism development. Beginning later provides richer covariate coverage but gives the model fewer pre-shock observations. The 2014 start balances those concerns and produces an estimate close to the center of the calibration distribution. The estimated passenger decline remains negative across the calibration range, and the preferred 2014 specification yields an estimated cumulative decline of about 538,000 departing passengers, very similar to the main estimate. All other hyperparameters follow the defaults recommended by Brodersen et al. (2015) unless otherwise noted.

Supplemental Figure S3 reports posterior inclusion probabilities for the primary passenger model. Each bar indicates how often a covariate is selected across posterior draws. Higher posterior inclusion probabilities identify indicators that are more consistently useful in predicting Iceland's pre-shock passenger trajectory. The figure also reports coefficient information, so it should be read as both a selection diagnostic and a sign/magnitude check. A

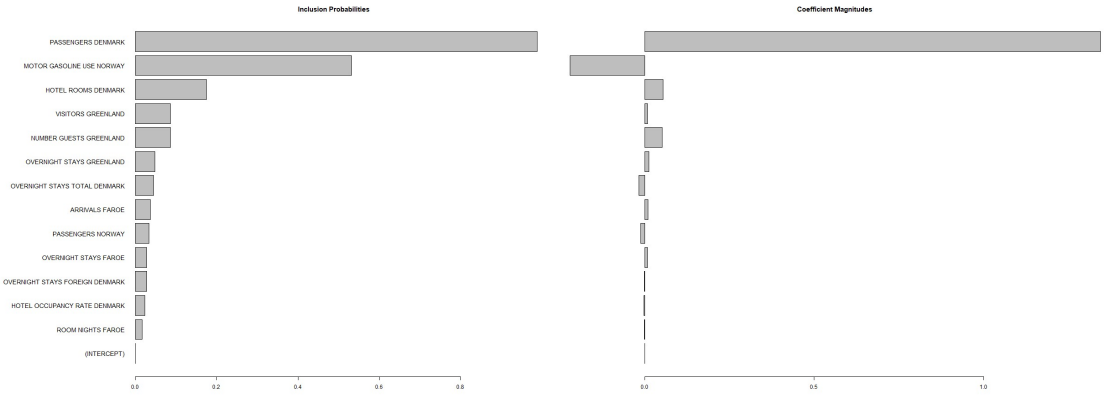
Figure S2: Cumulative Passenger Effects Across Alternative Training Windows



Note: Each point reports the cumulative April–December 2019 passenger effect estimated using a different pre-period start date. The median cumulative change across tested windows is approximately -458,000 passengers, the mean is approximately -487,000 passengers, and the 2014 specification is approximately -538,000 passengers.

high inclusion probability with a small coefficient can indicate a stable but modest predictor; a lower inclusion probability with a larger coefficient can indicate a predictor that is useful in some posterior draws but not consistently selected. This selection process helps address overfitting because weak predictors are shrunk toward zero rather than being forced into the counterfactual.

Figure S3: Posterior Inclusion Probabilities for the Passenger Model



Note: Posterior inclusion probabilities indicate how often each candidate predictor is selected by the spike-and-slab prior during estimation. Predictors with higher values contribute more consistently to the pre-shock counterfactual fit.

S2 Placebo Checks

Placebo checks test whether the model detects effects where none should be expected. Table S2 reports three categories of placebo tests. The first panel applies the bankruptcy timing to an unrelated Icelandic outcome, fish landings. The second applies the treatment timing to other countries. The third applies the treatment to earlier Icelandic periods before the bankruptcy.

The placebo results are reassuring because the model does not produce comparable effects in places or outcomes where the bankruptcy should not have had a direct impact. The fish-landings placebo is statistically insignificant, indicating that the model does not generate a false effect for an Icelandic sector unrelated to international air access. Placebos applied to Greenland, Denmark, and Norway are also statistically insignificant. The Faroe Islands estimate is larger in magnitude but statistically imprecise, consistent with volatility in a very small passenger market. The 2018 placebo is negative and weakly significant, which may reflect a cooling of Icelandic tourism before the bankruptcy. That pre-period placebo is important because it cautions against attributing every element of the 2019 slowdown to WOW Air. However, it is smaller and provides much weaker evidence against the null than the 2019 bankruptcy effect, and the preferred model is trained through January 2019 so that information from the pre-bankruptcy cooling period contributes to the counterfactual.

S3 Robustness Checks

The robustness checks vary the control-country set, covariate grouping, and time-series structure of the model. The purpose is to evaluate whether the passenger result depends on a particular modeling choice.

Table S2: Placebo Counterfactual Tests

Indicator	Relative Impact (%)	9-Month Cumulative Impact	95% CI Upper	95% CI Lower	p-value	RMSE (priors)
Total Passengers	-20.63***	-533,643	-251,386	-942,546	0.0002	13,530
Icelandic Passengers	-16.70***	-100,771	-23,111	-180,957	0.0010	2,905
Foreign Passengers	-22.75***	-461,168	-211,729	-831,090	0.0002	11,895
Fish Landings	-2.88	-32,226,026	135,201,145	-187,612,889	0.3374	23,174,309
Greenland	-0.14	-1,888	19,389	-26,211	0.4369	229
Faroe Islands	8.97	23,269	68,348	-33,982	0.1401	NA
Denmark	-3.20	-449	432	-1,285	0.1606	41
Norway	1.85	166,696	744,650	-555,206	0.2694	21,952
Impact in January 2018	-7.48*	-198,384	79,105	-454,292	0.0742	11,081
Impact in January 2017	8.32	146,501	486,210	-438,857	0.2444	11,023

Note: The first column lists each placebo or treated series. Relative impact is the estimated post-treatment deviation from the counterfactual as a percentage of the counterfactual. The cumulative-impact column reports the total post-treatment difference in original units. Placebo effects that are small or statistically insignificant support the interpretation that the main Iceland passenger effect is not a general modeling artifact. Stars denote statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

S3.1 Control-Country Selection

Table S3 omits one or more comparison countries from the preferred passenger model. The estimated passenger decline remains large and statistically significant across all specifications. The relative impact ranges from about -20% to -29%, with an average of -23.4%.

Table S3: Passenger Effects Under Alternative Control-Country Sets

Omitted Control Country	Relative Impact (%)	9-Month Cumulative Impact	95% CI Upper	95% CI Lower	p-value	RMSE (priors)
Greenland	-20.80***	-541088	-246711	-945642	4e-04	13870
Faroe Islands	-22.28***	-592873	-269139	-1088169	2e-04	12945
Denmark	-27.98***	-791612	-477276	-1092558	2e-04	19983
Norway	-20.38***	-531946	-214080	-1013803	8e-04	14075
Faroe Islands & Greenland	-21.47***	-561587	-255464	-956970	2e-04	13839
Greenland & Denmark	-28.77***	-820965	-537468	-1125084	2e-04	20322
Denmark & Faroe Islands	-21.15***	-545757	-272972	-817773	2e-04	22257
Denmark & Norway	-26.30***	-727234	-420244	-1024210	2e-04	23640
Norway & Faroe Islands	-20.97***	-552702	-217453	-1013604	2e-04	13780
Mean	-23.34***	-629529	-323423	-1008646	3e-04	17190

Note: Each row reports the main passenger model after omitting the listed control country or countries. The stability of the negative effect indicates that the main result is not driven by a single comparison economy. Stars denote statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

S3.2 Covariate Grouping

Table S4 varies the topic area of indicators used to estimate the counterfactual. The first column lists the covariate group. The estimated passenger decline remains statistically significant across all groups. The preferred tourism-indicator model balances substantive relevance and pre-period fit, but the direction and approximate scale of the effect are robust to broader or narrower predictor sets.

Table S4: Passenger Effects Under Alternative Covariate Groups

Indicator	Relative Impact (%)	9-Month Cumulative Impact	95% CI Upper	95% CI Lower	p-value	RMSE (priors)
All Indicators	-22.30***	-583478	-359199	-801125	2e-04	11422
Passenger Indicators	-17.56***	-435721	-210164	-692980	4e-04	15277
Tourism Indicators	-20.59***	-533643	-246692	-945415	2e-04	13530
Labor Indicators	-22.15***	-578100	-350623	-792190	2e-04	11089
Other Indicators	-23.69***	-630521	-361031	-896049	2e-04	26532
Mean	-21.26***	-552293	-305542	-825552	2e-04	15570

Note: Each row reports the passenger model estimated with a different group of candidate predictors. Relative impact is the estimated April–December 2019 passenger deviation from the counterfactual divided by the counterfactual. Stars denote statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

S3.3 Counterfactual Without Time-Series Components

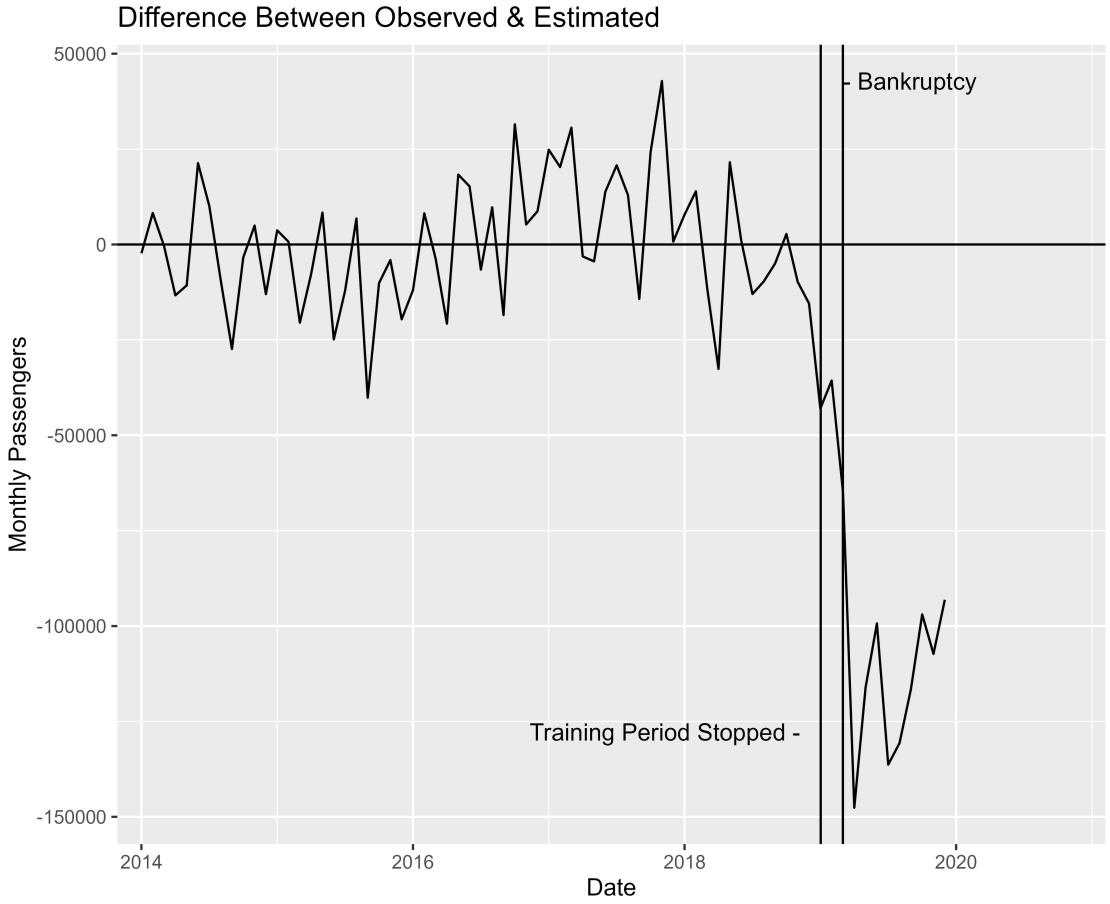
Table S5 compares the preferred Bayesian structural time-series model with a simplified regression that excludes time-series and seasonality components. The simplified model produces a larger estimated effect but a worse pre-period fit. This indicates that the preferred model is more conservative and better suited to a tourism outcome with strong seasonality and a changing trend.

Table S5: Passenger Effects With and Without Time-Series Components

Model Used	Loss Percent	Monthly Impact	Cumulative Impact	P-Value	RMSPE
Simple OLS Regression, Time Series Omitted	-34.01%***	-115,986	-1,043,878	5.2e-05	17,204
BSTS Preferred Model	-20.63%***	-59,294	-533,643	2e-04	13,530

Note: The first row reports a simplified regression using contemporaneous comparison indicators without local trend or seasonal components. The second row reports the preferred Bayesian structural time-series specification. RMSPE is the pre-period prediction error. Stars denote statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure S4: Monthly Passenger Differences Without Time-Series Components



Note: The figure reports monthly differences between observed departures and a simplified counterfactual that excludes time-series components. The simplified model produces larger errors than the preferred specification.

Figure S5: Cumulative Passenger Differences Without Time-Series Components



Note: The figure reports cumulative differences between observed departures and a simplified counterfactual that excludes time-series components.

S4 Two-Way Fixed-Effects Comparison

As a benchmark, a two-way fixed-effects model is estimated using log arrivals as the outcome:

$$\log(Y_{it}) = \beta_0 + \beta_1 Treat_i + \beta_2(Post_t \times Treat_i)$$

$$+\beta_3 CountryFE_i + \beta_4 YearFE_t + \beta_5 MonthFE_t + \epsilon_{it}.$$

The parallel-trends assumption is unlikely to hold because Iceland's tourism sector grew rapidly before the bankruptcy. Table S6 confirms this concern. Estimated treatment effects for Iceland vary sharply with the length of the pre-period and become positive when more years are included. Some control-country placebos also become statistically significant. These patterns support the use of the Bayesian structural time-series model in the main analysis.

Table S6: Two-Way Fixed-Effects Treatment Estimates Across Pre-Period Windows

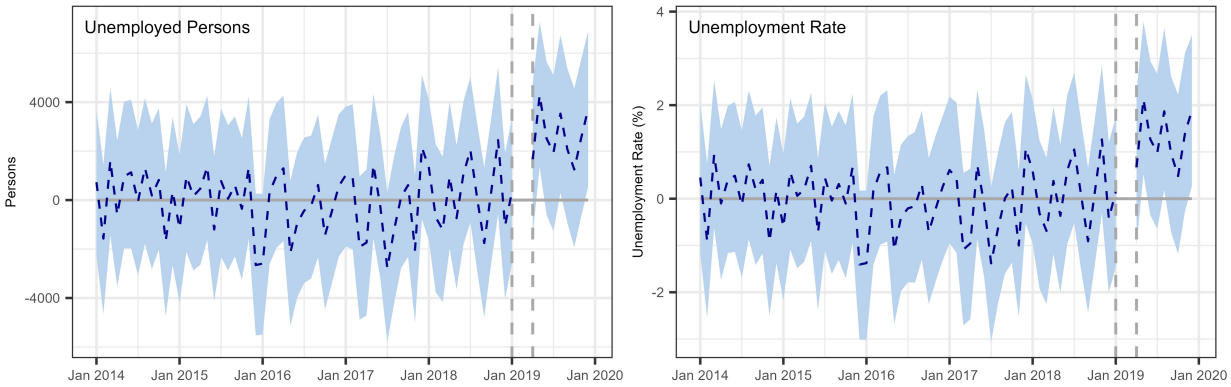
	1 year prior	2 years prior	3 years prior	4 years prior	5 years prior	6 years prior
Iceland	-0.2177*	-0.1527**	-0.0641	0.0493	0.1494**	0.2396***
Greenland	0.1395	0.03	-0.0128	-0.0438	-0.0629	-0.0898
Faroe Islands	0.2512**	0.1861***	0.2032***	0.1919***	0.1885***	0.1832***
Norway	-0.0706	-0.0245	-0.0551	-0.1054*	-0.158**	-0.1981***
Denmark	-0.1025	-0.0389	-0.0712	-0.0919*	-0.117*	-0.1349**

Note: Each cell reports the estimated interaction between the treated country and the post-bankruptcy period using a different pre-period length. Rows identify the country assigned to treatment. Instability across pre-period windows and significant placebo effects indicate that two-way fixed effects are poorly suited to this setting. Stars denote statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

S5 Additional Outcome Tables and Figures

Figures S6 and S7 show additional labor outcomes during the post-bankruptcy months. The first compares national unemployment measures with their counterfactual paths. The second shows monthly differences between observed and counterfactual labor levels across selected outcomes, with shaded intervals indicating uncertainty.

Figure S6: National Unemployment Effects of the WOW Air Bankruptcy



Note: Panels report observed unemployment outcomes relative to Bayesian structural time-series counterfactuals for April–December 2019. Unemployed persons are measured in persons; the unemployment rate is measured in percentage points.

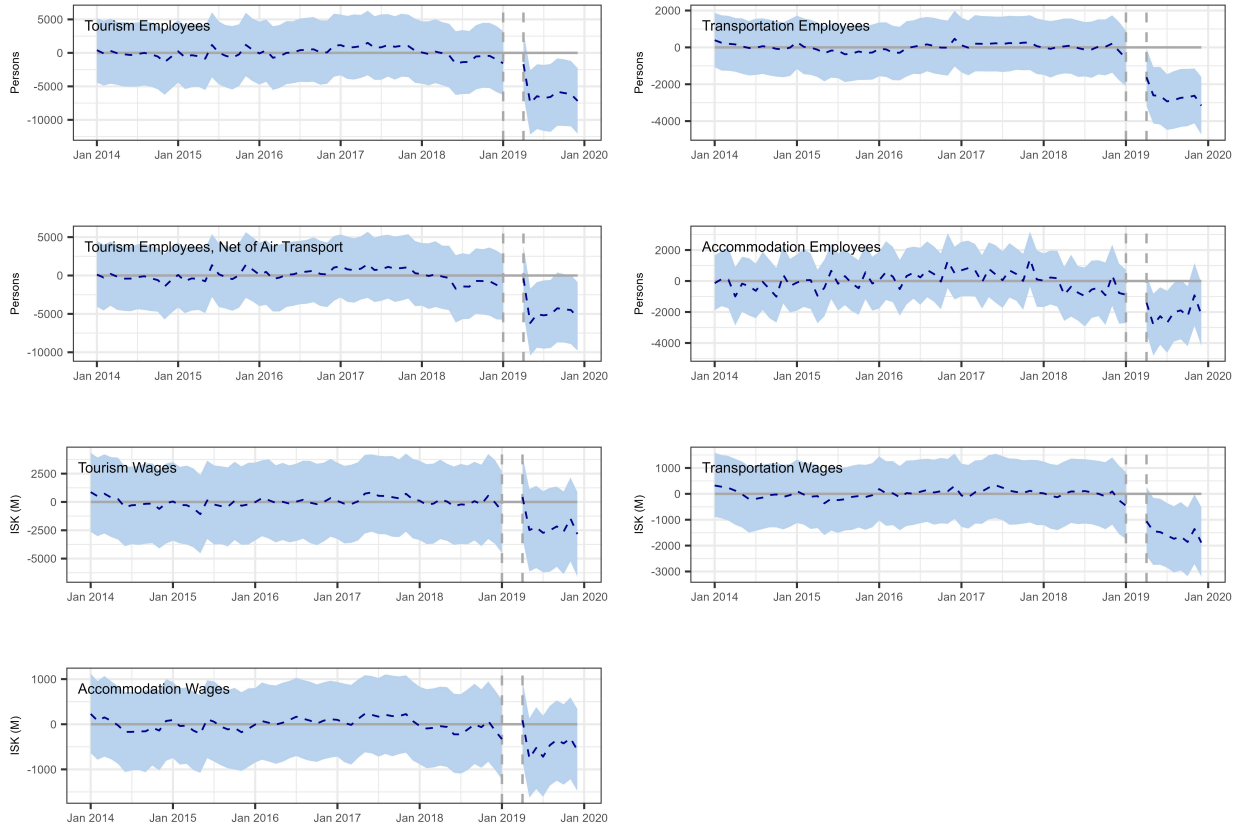
Table S7 reports additional activity outcomes connected to visitor movement and local tourism demand. The rows use the same counterfactual setup as the main tables and estimate effects for capital-area road traffic, registered rental cars, and overnight stays.

Table S7: Additional Tourism Activity Effects of the WOW Air Bankruptcy

Outcome	Relative Impact (%)	Avg. Monthly Impact	Monthly Avg. Predicted	Monthly Avg. Observed	YoY Actual Change (%)	p-value	RMSE (pre-period)
Capital-Area Road Traffic	-8.48***	-4,073	47,637	43,564	0.99	0.0051	822
Registered Rental Cars	-20.88***	-6,653	31,499	24,846	-3.63	0.0004	969
Overnight Stays	-4.28	-18,325	410,358	392,033	2.34	0.1680	19,523

Note: Rows report counterfactual estimates for Icelandic visitor-adjacent activity measures. Capital-area road traffic measures road traffic in the capital region; registered rental cars measures the rental-car stock; overnight stays measures overnight stays in Iceland. Average monthly impact is measured in the original units of each row. Stars denote statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure S7: Monthly Differences Between Observed and Counterfactual Labor Levels



Note: Lines report monthly differences between observed labor outcomes and their Bayesian structural time-series counterfactuals. Shaded areas indicate posterior uncertainty.